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CSA0311

Data Structures

Co1 AT3

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     printf("Enter size: ");
6     scanf("%d", &n);
7     int arr[n];
8     printf("Enter %d elements: ", n);
9     for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
10
11    for (int i = 0, j = n - 1; i < j; i++, j--) {
12        int t = arr[i];
13        arr[i] = arr[j];
14        arr[j] = t;
15    }
16
17    printf("Reversed: ");
18    for (int i = 0; i < n; i++) printf("%d ", arr[i]);
19    printf("\n");
20    return 0;
21 }
```

Submitted

Input

5
1 2 3 4 5

Output



Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     char a[100];
5     int i = 0, j, p = 1;
6     char ch;
7
8     while ((ch = getchar()) != '\n' && ch != EOF) {
9         a[i] = ch;
10        i++;
11    }
12
13    j = i - 1;
14    i = 0;
15
16    while (i < j) {
17        if (a[i] != a[j]) {
18            p = 0;
19        }
20        i++;
21        j--;
22    }
```

Input

MADAM

Output



Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, key, low, high, mid, i;
5
6     printf("Enter size: ");
7     scanf("%d", &n);
8
9     printf("Enter sorted array elements: ");
10    for(i = 0; i < n; i++)
11        scanf("%d", &a[i]);
12
13    printf("Enter element to search: ");
14    scanf("%d", &key);
15
16    low = 0;
17    high = n - 1;
18
19    while(low <= high) {
20        mid = (low + high) / 2;
21
22        if(a[mid] == key) {
23            printf("Element found at position %d", mid + 1);
```

Input

```
5
10 20 30 40 50
30
```

Output



Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, key, i, count = 0;
5
6     printf("Enter size: ");
7     scanf("%d", &n);
8
9     printf("Enter array elements: ");
10    for(i = 0; i < n; i++)
11        scanf("%d", &a[i]);
12
13    printf("Enter element to count: ");
14    scanf("%d", &key);
15
16    for(i = 0; i < n; i++) {
17        if(a[i] == key)
18            count++;
19    }
20
21    printf("Occurrences of %d = %d", key, count);
22
23    return 0;
```

Input

```
6
2 5 2 8 2 4
2
```

Output



Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, key;
5     int low, high, mid, i;
6
7     printf("Enter size: ");
8     scanf("%d", &n);
9
10    printf("Enter sorted array elements: ");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter element to search: ");
15    scanf("%d", &key);
16
17    low = 0;
18    high = n - 1;
19
20    while(low <= high) {
21        mid = (low + high) / 2;
22
23        if(a[mid] == key) {
```

Input

```
5
10 20 30 40 50
40
```

Output



Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, key;
5     int low, high, mid, i;
6
7     printf("Enter size: ");
8     scanf("%d", &n);
9
10    printf("Enter sorted array elements: ");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter element to search: ");
15    scanf("%d", &key);
16
17    low = 0;
18    high = n - 1;
19
20    while(low <= high) {
21        mid = (low + high) / 2;
22
23        if(a[mid] == key) {
```

Input

```
5
10 20 30 40 50
40
```

Output



Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[50], b[50], c[100];
5     int n1, n2, i, j;
6
7     printf("Enter size of first array: ");
8     scanf("%d", &n1);
9
10    printf("Enter first array elements: ");
11    for(i = 0; i < n1; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter size of second array: ");
15    scanf("%d", &n2);
16
17    printf("Enter second array elements: ");
18    for(i = 0; i < n2; i++)
19        scanf("%d", &b[i]);
20
21    for(i = 0; i < n1; i++)
22        c[i] = a[i];
23}
```

Input

```
3
1 2 3
3
4 5 6
```

Output



Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, i, largest, smallest;
5
6     printf("Enter size: ");
7     scanf("%d", &n);
8
9     printf("Enter array elements: ");
10    for(i = 0; i < n; i++)
11        scanf("%d", &a[i]);
12
13    largest = smallest = a[0];
14
15    for(i = 1; i < n; i++) {
16        if(a[i] > largest)
17            largest = a[i];
18
19        if(a[i] < smallest)
20            smallest = a[i];
21    }
22
23    printf("Largest Element = %d\n", largest);
24}
```

Input

```
5
10 45 7 23 18
```

Output



Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, i;
5     int largest, second;
6
7     printf("Enter size: ");
8     scanf("%d", &n);
9
10    printf("Enter array elements: ");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    largest = second = a[0];
15
16    for(i = 1; i < n; i++) {
17        if(a[i] > largest) {
18            second = largest;
19            largest = a[i];
20        }
21        else if(a[i] > second && a[i] != largest) {
22            second = a[i];
23        }
24    }
```

Input

```
5
10 45 20 30 15
```

Output



```

1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, key, i;
5     int first = -1, last = -1;
6
7     printf("Enter size: ");
8     scanf("%d", &n);
9
10    printf("Enter array elements: ");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter element to search: ");
15    scanf("%d", &key);
16
17    for(i = 0; i < n; i++) {
18        if(a[i] == key) {
19            if(first == -1)
20                first = i;
21            last = i;
22        }
23    }

```

```

6
2 5 3 5 8 5
5

```

Output



Your Code

```

1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, i, sum = 0, total, missing;
5
6     printf("Enter number of elements: ");
7     scanf("%d", &n);
8
9     printf("Enter array elements: ");
10    for(i = 0; i < n; i++) {
11        scanf("%d", &a[i]);
12        sum += a[i];
13    }
14
15    total = (n + 1) * (n + 2) / 2;
16    missing = total - sum;
17
18    printf("Missing Element = %d", missing);
19
20    return 0;
21 }

```

Input

```

5
1 2 3 5 6

```

Output

